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Syllabus - CSC252 - S25

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CSc 252 - Computer Organization - Spring 25

5:00 - 6:15 pm, TuTh, Gittings 201

Course Description

Basic machine organization; elementary hardware concepts; CPU internals. Machine operations and instructions; assembly language concepts and programming.

Instructor and Contact Information

Name: Diana Diazh, Ph.D.
Email: ddiana@arizona.edu
Office: [Gould-Simpson](#) 835

Teaching Assistants: This class has several teaching assistants. Their contact information and office hours can be found on the class website.

Office Hours

Exact office hours are subject to change. Consult the class webpage for up-to-date information.

Websites

Class Website: The main course website will host the majority of the class content, including the syllabus, slides, lecture notes, and assignments. The class website is <https://profdiazh.github.io/csc252/>

D2L

We will use D2L for posting grades <https://d2l.arizona.edu/d2l/home/1550704>

Discord

We will use Discord as the primary platform for announcements, discussion, and Q&A. If you have a general question, post it publicly. If you have a private question, or you would like to include your work or solutions to a problem or assignment, DM or email me or your TAs. Never include your code or solutions in a public post.

All students must sign up for the Discord server here <https://discord.gg/ZDBUsGpGvS>

Course Format and Teaching Methods



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

you must work diligently with your table and arrive at the best answer you can. Ask questions early and often to maximize your success in this class!

Course Objectives

Key topics:

- Binary numbers and arithmetic
- Assembly language programming: common concepts and their implementation in MIPS
- Functions, data structures, and the memory model in assembly
- Simple hardware implementations, starting at the gate level and working upwards
- Simple CPU design, including ALUs, instruction decode, and basic pipelining

Expected Learning Outcomes

By the end of the semester, the student will:

- Demonstrate an understanding that a computer is made up entirely of a set of bits, and understand how types of variables are encoded and arranged in memory
- Demonstrate an understanding of how computational operations (such as addition, comparison, or branching) can be implemented using simple electrical devices
- Demonstrate an understanding of how higher-level programming constructs (variables, data structures, functions, etc.) are represented and operated
- Demonstrate the ability to think of a computer program as a finite set of time-discrete steps, each of which is made up of extremely simple operations, and be able to explain how these small steps can be used to implement complex, multi-cycle calculations.

Transferable Career Skills

- Technology: Students will use technology to investigate how the computer stores data and execute tasks.
- Teamwork: The course will include lectures, in-class discussions, and activities emphasizing in-class teamwork.
- Communication: Students are encouraged to emphasize communication by interacting with teaching assistants and using various channels, such as D2L, in-class discussions, and Discord, to stay updated on course materials.

Makeup Policy for Students Who Register Late

Students who register for class late must email Prof. Diazh about their late registration to be allowed to make up missed assignments; all missed assignments will be



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Syllabus - CSC252 - S25

Updated automatically every 5 minutes

class Discord server, the link to which is available through D2L and the class website. If a student has an issue that cannot reasonably be resolved through Discord (inappropriate to discuss publicly, private issue, etc.), they may email or DM their TA or the instructor.

Required Texts or Readings

[Required] To access the online textbook, please log in to [d2l](#) → go to Content → Pay One Price and Inclusive Access Materials → Textbook Access to get your access code. Then, go to Content → Week 2 and follow the steps to set up an account. The textbook will be the online version of: *Patterson and Hennessy, Computer Organization and Design, 5th Ed. Morgan Kaufmann*.

The instructor may assign additional readings from online articles, blogs, papers, and library resources.

Pay One Price: The program from the UA Campus Store provides required textbooks and courseware for a flat rate each semester. If you're registered for a Main Campus undergraduate course, you're automatically enrolled in Pay One Price each semester (fall & spring & summer per month). You can opt out via D2L to receive a refund by the refund deadlines posted on

https://shop.arizona.edu/payoneprice_student

Required or Special Materials

It is recommended that each student have a laptop or tablet available for use during in-class activities and assignments. If you do not own a portable laptop, you can borrow one from the University library: <https://lib.arizona.edu/borrow/tech>

Required Extracurricular Activities

None

Assignments and Examinations: Schedule/Due Dates

Projects

This class will have programming projects, which are due most weeks. The following schedule gives approximate due dates for each one. The schedule below is approximate; as I assign the projects, it may occasionally be necessary to adjust due dates. Likewise, the topics mentioned are my current plans and are subject to change. It is possible that I might assign fewer projects than listed below; however, I will not assign more.

Projects will be due at 7 pm, but TAs are not allowed to answer any questions from



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Syllabus - CSC252 - S25

Updated automatically every 5 minutes

Assembly Project 1	Wednesday, February 5	Introduction to MIPS
Simulation Project 2	Wednesday, February 12	Logic Networks
Assembly Project 2	Wednesday, February 19	Loops, Arrays, and Strings
Simulation Project 3	Wednesday, February 26	An ALU
Assembly Project 3	Wednesday, March 5	Intro to Functions
Simulation Project 4 - Milestone 1	Wednesday, March 19	Instruction Decode
Simulation Project 4 - Milestone 2	Wednesday, March 26	Single-Cycle CPU
Assembly Project 4	Wednesday, April 2	
Assembly Project 5	Wednesday, April 9	
Simulation Project 5	Wednesday, April 16	Pipelined CPU
Assembly Project 6	Wednesday, April 23	
Simulation Project 6	Wednesday, April 30	No Project

Late Days

This class offering does not allow any Late Days.

Tests

This class will have 6 short tests. Each will be given during the lecture and take less than the full class period (30 minutes); after the test is complete, we will continue lecturing. Tests will be given on the following days:

Test 1	Thursday, February 6
Test 2	Thursday, February 20
Test 3	Thursday, March 6
Test 4	Thursday, March 27
Test 5	Thursday, April 10
Test 6	Thursday, April 24

Students with DRC accommodations can take the test at the DRC location on the same date as the test at 4:30 pm or 4:00 pm, depending on their accommodations.



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

inspect them to make sure that the students completed the assigned problems but will not grade them for correctness. Homework is intended to be test preparation; as such, students are encouraged to work in groups. However, every student must turn in a paper copy of the homework; shared papers will not be accepted. The homework will not include all the topics asked in the test, but the majority of them.

zyBook Readings

Most weeks, a few sections from the Zybooks textbook will be assigned. The zyBooks website automatically checks your answers; the sections must all be completed by Monday, before the first lecture that week. We will collect completion information from the website and give points based on that information.

Your points assignments will be based on the percentage of the reading you completed. The intent is for the Lecture to be a time when we explain and expand on the textbook - instead of it being your first time to see each topic.

Readings are due at midnight (11:59 pm MST). The planned due dates for the various readings are:

Reading title	Deadline
Week 2	Monday, January 20
Week 3	Monday, January 27
Week 4	Monday, February 3
Week 5	Monday, February 10
Week 6	Monday, February 17
Week 7	Monday, February 24
Week 8	Monday, March 3
Week 9	Monday, March 17
Week 10	Monday, March 24
Week 11	Monday, March 31
Week 12	Monday, April 7
Week 13	Monday, April 14
Week 14	Monday, April 21
Week 15	Monday, April 28

Note: Make-up exams will not be given. At the end of the semester, I will replace your



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

that would cause you to miss a test.

Please note that should you miss multiple tests under sub-extreme circumstances, you will definitely get a zero for those additional missed tests.

Students with DRC accommodations should request to be scheduled to take their test between 2 pm and 5:15 pm the same day of the test so they can attend the last 45 minutes of the lecture and earn participation points.

Final Examination

The Final exam will run 2 hours (120 minutes). Other than its length, its format will be similar to the 6 previous tests. As per university policy, the final exam will be on **Wednesday, May 14, from 6:00 to 8:00 pm** in our regular classroom.

University Policies Regarding Finals:

Final Exam Regulations and Final Exam Schedule,

<https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams>

Grading Scale and Policies

Grading Scale

I will use a simple grade cutoff scheme. This means that if you earn the number of points listed for a given grade, you are guaranteed that grade. At the end of the semester, I reserve the right to lower these cutoffs, meaning that it might be easier to earn a good grade (but it is unusual for me to do so). However, I guarantee that I will not raise these cutoffs (making it harder to earn a good grade).

- 90% A
- 80% B
- 70% C
- 60% D

Point Distribution

Points will be distributed as follows:

			points per item	total points
5%	zyBooks Readings	drop 1 week	3.6	50
24%	Assembly Projects	drop 1 asm	48	240
21%	Simulation Projects	drop 1 sim	35	210



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

	Inclass Activities	Extra Credit	1	24
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Within each category, points are distributed evenly; that is, every Test is worth the same as every other (except the Final), every Assembly Project is worth the same as every other, and every Simulation Project is worth the same as every other. Thus, the exact value of each item will depend on the number of each item that is assigned.

Grading Schedule

Projects will be graded, typically, within 7 days of the due date. If exceptions have to be made occasionally, staff will inform the students about the delay and the reason for it.

Tests will be graded within 10 days.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at <https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>.

Dispute of Grade Policy:

If you have an issue with how any item was graded, you must submit a request for a regrade within 7 days of when the item was returned to you. For the Tests, please use GradeScope's "Regrade Request" feature; for other items, email your TA. If you and your TA cannot come to an agreement, you may email me, but please use the ordinary path first.

Honors Credit

Students wishing to contract this course for Honors Credit should e-mail me to set up an



Syllabus - CSC252 - S25

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The UA's policy concerning Class Attendance, Participation, and Administrative Drops is available

at <https://catalog.arizona.edu/policy/class-attendance-and-participation>

The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable: <http://policy.arizona.edu/human-resources/religious-accommodation-policy>.

Absences pre-approved by the UA Dean of Students (or dean's designee) will be honored.

See <https://deanofstudents.arizona.edu/policies/attendance-policies-and-practices> -

Administrative Drops:

In past semesters, students have enrolled in CS classes but have not submitted any work, resulting in a grade of 'E' at the end of the term. To prevent this, I will administratively drop all students who have not submitted anything to Gradescope before the no-W drop date (1/28/25).

In addition, administrative drops will be in place throughout the semester to ensure your success in this course and that you are on top of the work that is due every week. Students who fail to submit assignments for two consecutive weeks will be administratively dropped. This is because students who miss coursework for two consecutive weeks or more often get an E at the end of the term in this course.

Scheduled Topic and Activities

For a list of assignment and test dates, please see the "Assignments and Examinations: Schedule/Due Dates" section above. The lectures will cover the following material. Note that this schedule is approximate and may need to be modified slightly during the semester.

Students are responsible for all material covered in the lecture (as well as the



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Syllabus - CSC252 - S25

Updated automatically every 5 minutes

Week 5	MIPS: loops and arrays
Week 6	ALUs
Week 7	MIPS: functions
Week 8	Simple CPU overview
Week 9	Instruction formats, simple CPU details
Week 10	Pipelining, part 1
Week 11	MIPS: recursion
Week 12	Pipelining, part 2: Hazards and forwarding
Week 13	Carry lookahead adders; multipliers
Week 14	Caches
Week 15	Floating point
Week 16	Review

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, playing games, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lectures. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave the lecture or discussion and may be reported to the Dean of Students.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Also, watch the video available at

https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spfurl=common%2Flearningeventdetail%2Fcrty0000000000003841

University-wide Policies link

Links to the following UA policies are provided here:

<https://catalog.arizona.edu/syllabus-policies>

- Absence and Class Participation Policies



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

principles and applications of course materials at a high level. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See <https://deanofstudents.arizona.edu/student-rights-responsibilities/academic-integrity>.

Uploading material from this course to a website other than Gradescope is strictly prohibited and will be considered a violation of the course policy and a violation of the code of academic integrity. Obtaining material associated with this course (or previous offerings of this course) on a site other than the course website, such as Chegg, Course Hero, StackOverflow etc., or any materials during a quiz or exam is a violation of the code of academic integrity. Any student determined to have uploaded or accessed material in an unauthorized manner will be reported to the Dean of Students for a Code of Academic Misconduct violation.

Generative AI use is NOT allowed for any purpose. In this course, any and all uses of generative artificial intelligence (AI)/large language model tools such as ChatGPT, Dall-e, Google Bard, Microsoft Bing, Copilot, etc. will be considered a violation of the Code of Academic Integrity, specifically the prohibition against submitting work that is not your own. This applies to all assessments in the course, including case studies, written assignments, discussions, quizzes, exams, and problem sets. This course policy is driven by the learning goals and desired learning outcomes for the course: The following actions are prohibited:

- entering all or any part of an assignment statement or test questions as part of a prompt to a large language model AI tool;
- incorporating any part of an AI-written response in an assignment;
- using AI to summarize or contextualize reading assignments or source materials; and
- submitting your own work for this class to a large language model AI tool for iteration or improvement.

The University Libraries have some excellent tips for avoiding plagiarism, available at <http://new.library.arizona.edu/research/citing/plagiarism>.

Sharing or selling class notes and/or other course materials to other students or a third party is not permitted without the instructor's express written consent. Violations of this and other course rules are subject to the Code of Academic Integrity and may result in course sanctions. Additionally, students who use D2L or UA e-mail to sell or buy these copyrighted materials are subject to Code of Conduct



Syllabus - CSC252 - S25

Updated automatically every 5 minutes

having a high similarity with any other source, you will need to meet and explain all your code to Prof. Diaz. Failing to demonstrate understanding of any concept, function, or tool used in your source code will be reported as an academic integrity violation to the Dean of Students. In other words, you must write by yourself and understand 100% of the code you submit to Gradescope.

Department-wide Syllabus Policies and Resources link

Links to the following departmental syllabus policies and resources are provided here: <https://www.cs.arizona.edu/cs-course-syllabus-policies> :

- Department Code of Conduct
- Illnesses and Emergencies, Obtaining Help, Preferred Names and Pronouns
- Confidentiality of Student Records
- Additional Resources, Land Acknowledgement Statement

Class Recording

Students should be mindful that instructor course content is subject to intellectual property protections and that fellow students have privacy rights and expectations as part of class activities. Some students may have an approved accommodation from the Disability Resource Center, which automatically notifies instructors through the DRC Instructor portal. Students who do not have a DRC accommodation must notify the instructor if they wish to record (audio and/or video) or photograph any class activity. When course activities are recorded, they should be used for a student's own personal educational use only.

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.

2/7/25